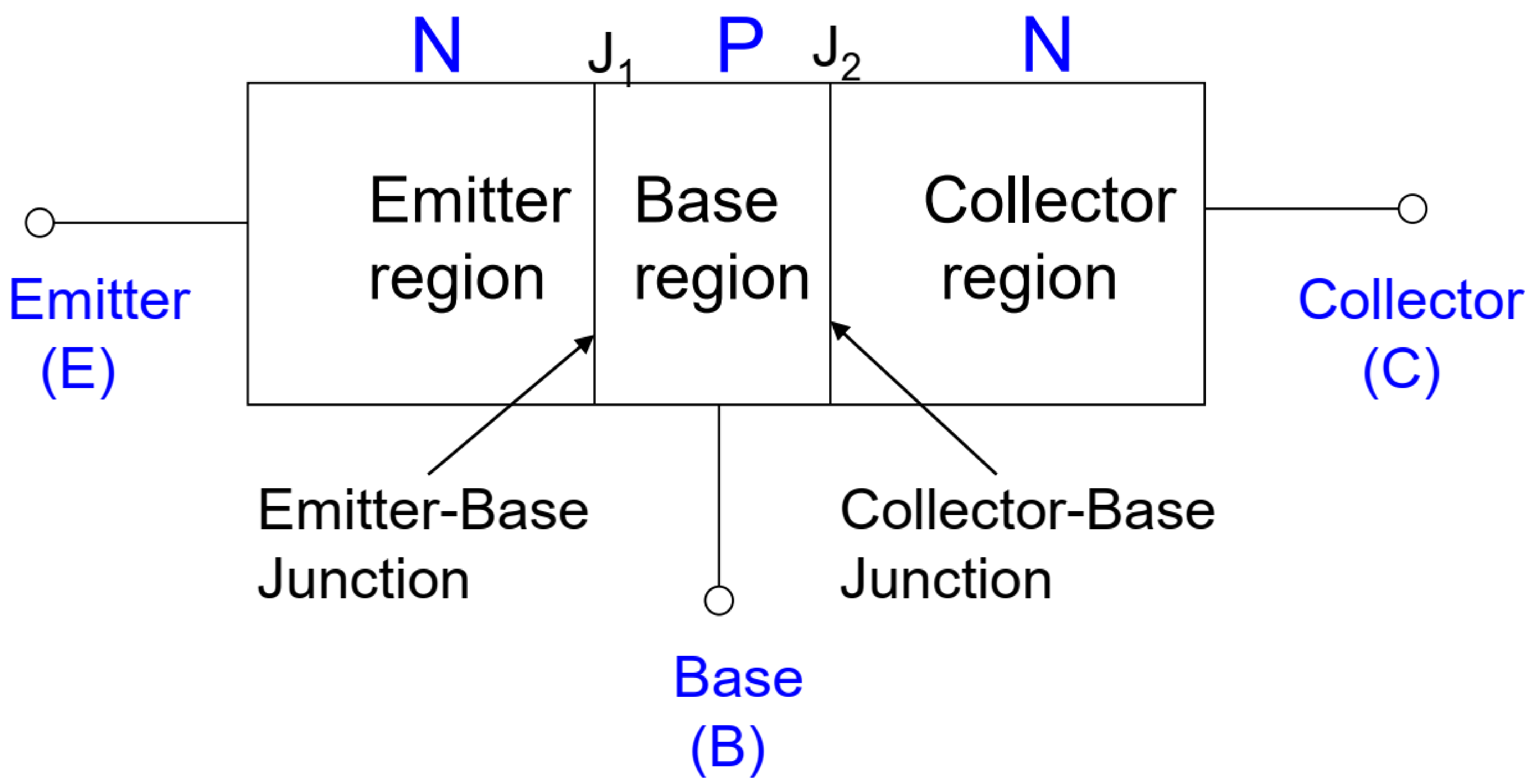
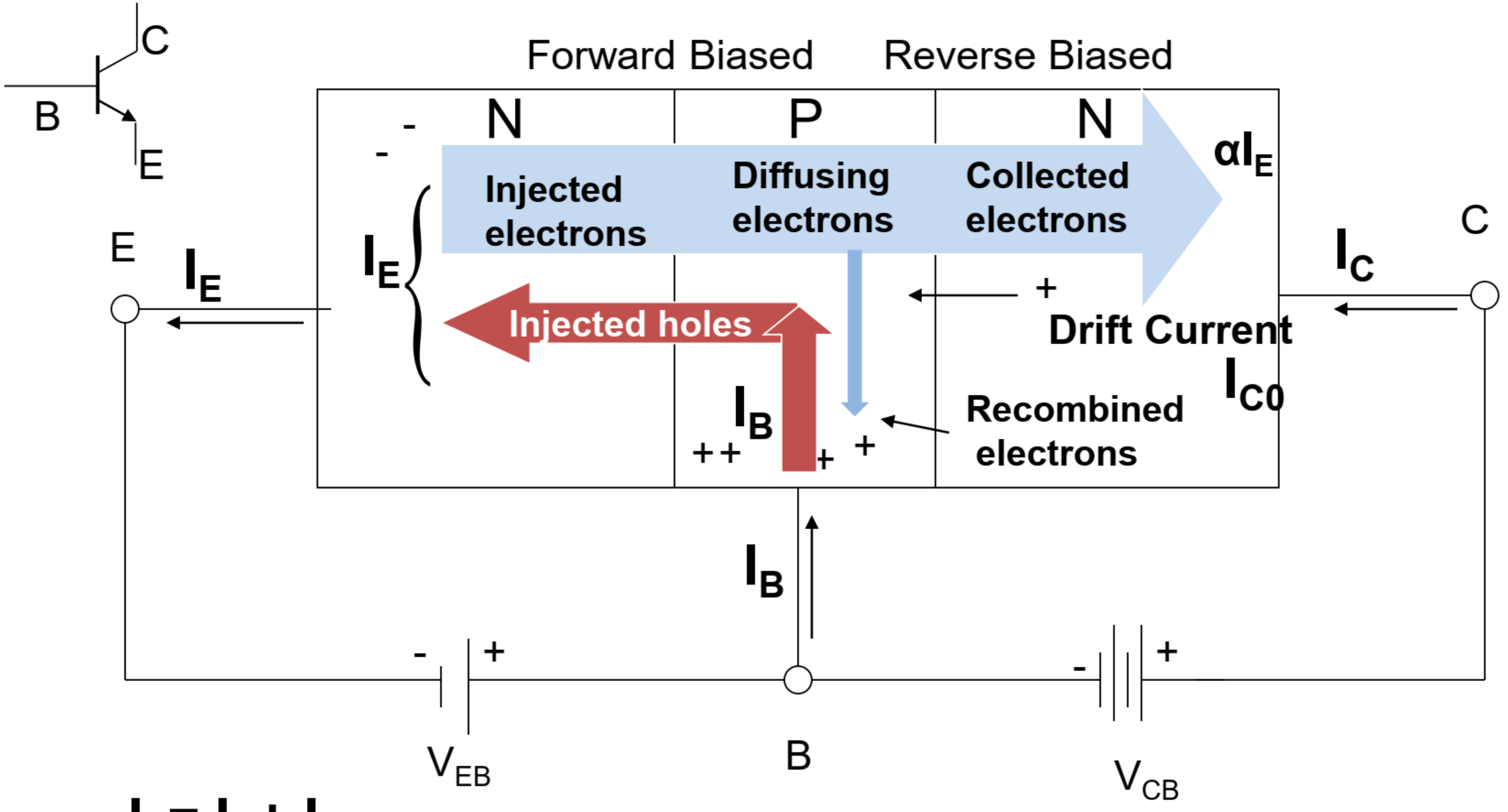
Bipolar Junction Transistor

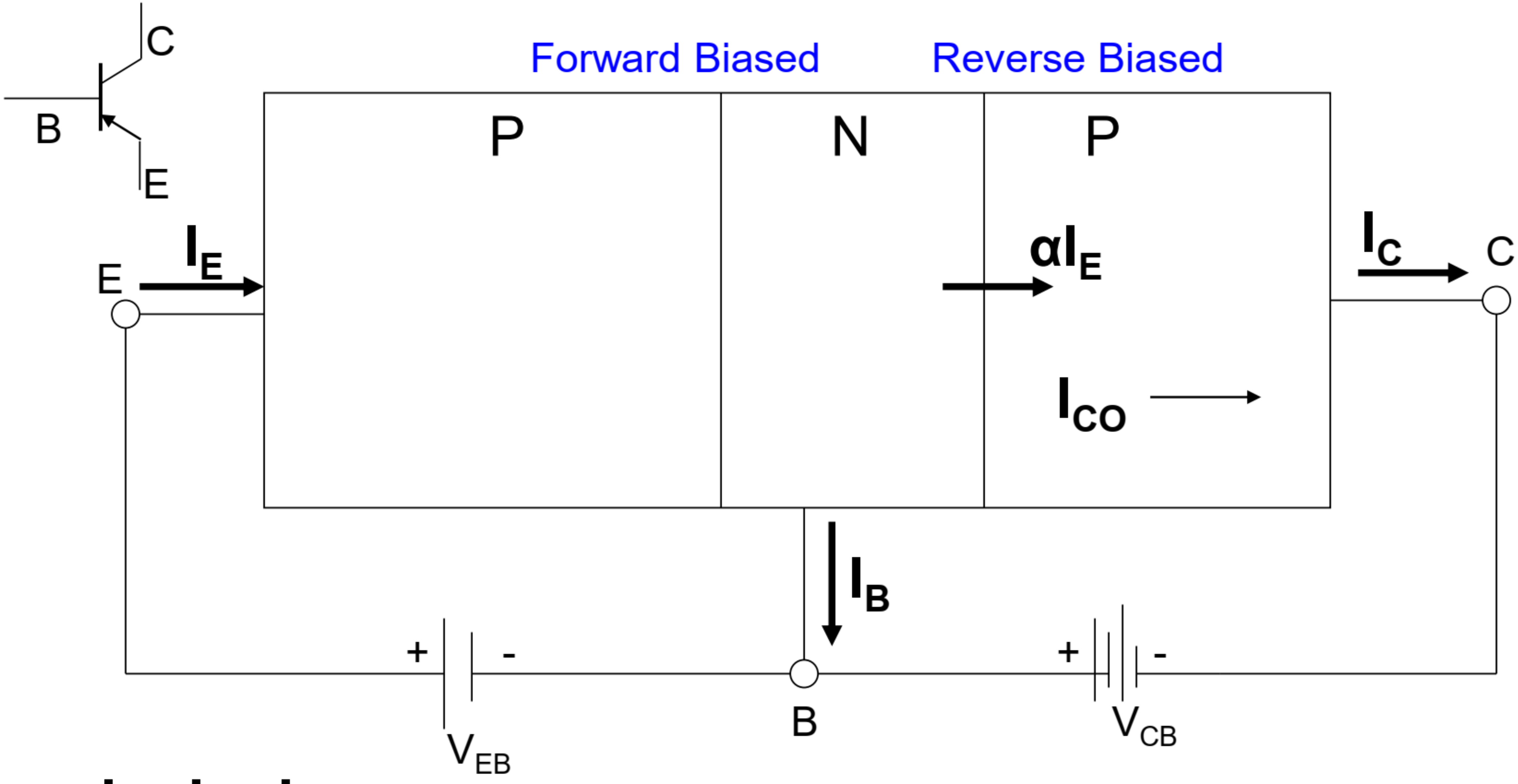

N
J1
P
J2
N
Emitter region
Base region
Collector region
Emitter (E)
Collector (C)
Emitter-Base Junction
Collector-Base Junction
Base (B)

Operation of an NPN Transistor in the Active mode



$I_E = I_B + I_C$

$I_C = \alpha I_E + I_{CO}$



$$I_E = I_B + I_C$$

$$I_C = \alpha I_E + I_{CO}$$

α is the fraction of the total emitter current which goes from the emitter, across the base, to the collector

$$I_C = \alpha I_E + I_{CO}$$

$$I_E = I_B + I_C$$

$$I_C = \alpha(I_B + I_C) + I_{CO}$$

$$(1-\alpha)I_C = \alpha I_B + I_{CO}$$

Transistor in Active Mode

$$I_C = \frac{\alpha}{1-\alpha} I_B + \frac{1}{1-\alpha} I_{CO}$$

Define $\beta = \dfrac{\alpha}{1-\alpha}$

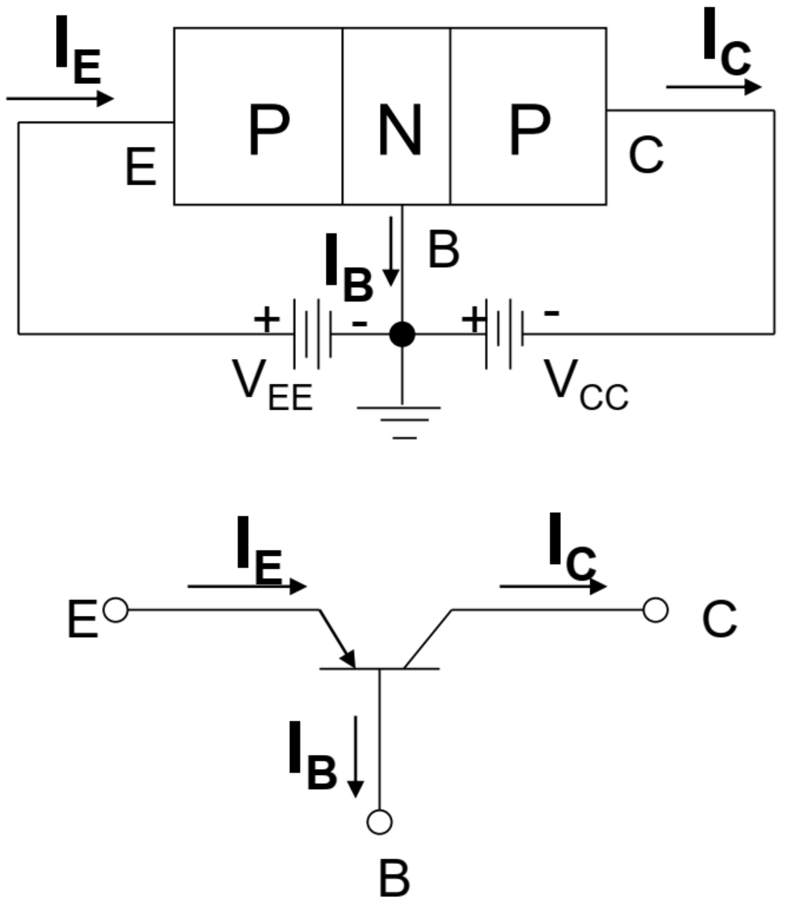
$$I_C = \beta I_B + (\beta + 1) I_{CO}$$

A Transistor may be operated in three different configurations :

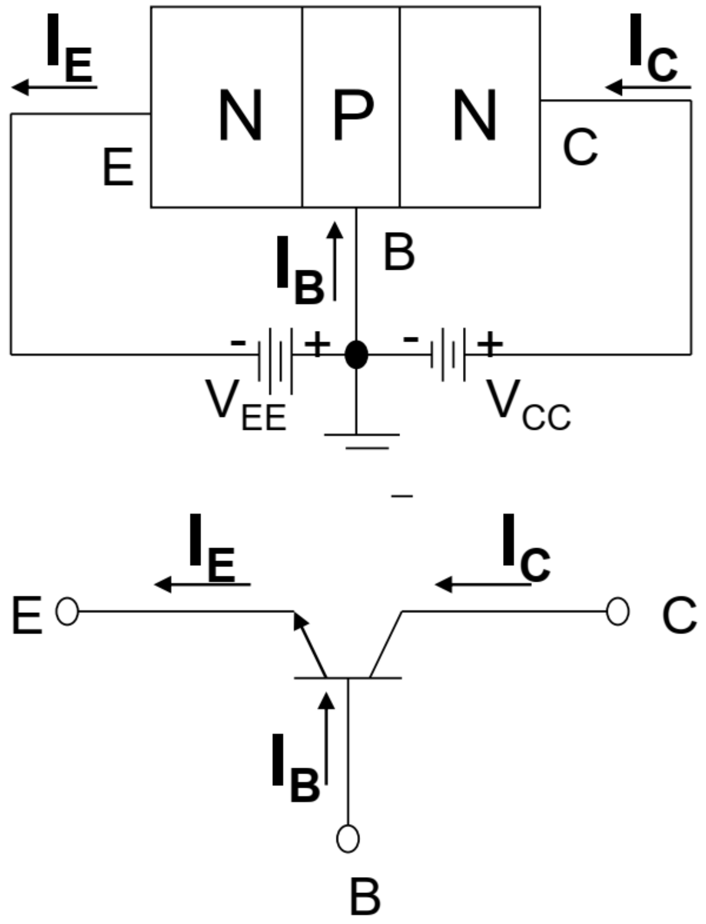
- Common Base (CB)
- Common Emitter (CE)
- Common Collector (CC)

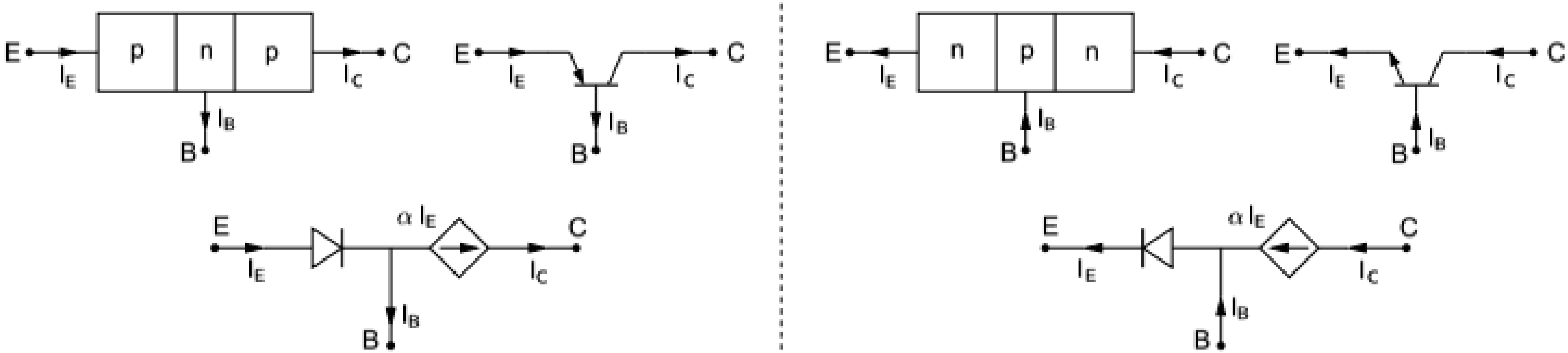
Common Base (CB) Transistor Configuration

PNP



NPN





E
C
B
p
n
p
I_E
I_C
I_B
α I_E
n
p
n

Common Emitter (CE) Transistor Configuration